Adaptive Multi-scale MCMC: Self-Tuning Alpha via Robbins-Monro Approximation

B.21 — Algorithm Design / Adaptive Search

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Abstract

Multi-scale MCMC (MultiScale) improves redistricting chain mixing by alternating between fine-level tract proposals and coarse-level county proposals, with a fixed probability α of making a coarse move at each step. The optimal α varies by state: Texas ($k = 38$, 254 counties, 5,265 tracts) benefits from $\alpha \approx 0.30$, while Iowa ($k = 4$, 99 counties, 825 tracts) benefits from $\alpha \approx 0.15$. Manual tuning of α across 50 states is impractical.

We introduce AdaptiveMS, which tunes α automatically using *Robbins-Monro stochastic approximation* [Robbins and Monro, 1951]. Every $\Delta = 50$ steps the algorithm measures the observed coarse acceptance rate in the current window; if it exceeds the target (0.30) α is increased; if below, α is decreased. The Robbins-Monro step size $\gamma_t = \gamma_0/\sqrt{t}$ satisfies the classical convergence conditions ($\sum \gamma_t = \infty$, $\sum \gamma_t^2 < \infty$), so the alpha trace converges to the state-specific optimal under a monotonicity assumption empirically satisfied for typical redistricting county graphs.

Empirically, AdaptiveMS converges: North Carolina ($k = 14$) stabilises at $\alpha^* \approx$ 0.28, Texas ($k = 38$) at ≈ 0.33, and Iowa ($k = 4$) at ≈ 0.17. Autocorrelation of the

edge-cut objective at lag 100 matches or improves on fixed-α MultiScale in all three states, with no manual tuning. AdaptiveMS is a drop-in replacement for `-search multiscale`, exposed as `-search multiscale-adaptive`.

Contents

1. Introduction

1.1. The Alpha Hyperparameter in Multi-scale MCMC

Multi-scale MCMC (MultiScale) is a Search-layer strategy for redistricting chains introduced in G.11 [Della-Libera, 2026]. The algorithm alternates between two Markov kernels: a *fine-level* ReCom move on the full census-tract graph, and a *coarse-level* move on the county-aggregated graph, projected back to the tract level via a rebalancing pass. Coarse moves allow the chain to escape fine-scale local optima by proposing large-scale boundary rearrangements at once; fine moves maintain detailed feasibility.

The key hyperparameter is $\alpha \in (0, 1)$: the probability of making a coarse move at each step. Setting α too low underuses the coarse level, leaving the chain effectively equivalent to standard ReCom. Setting α too high causes the chain to repeatedly propose coarse moves that fail the rebalancing check, wasting steps.

1.2. State-Specific Sensitivity

The optimal α is not universal. It depends on three structural properties of the state:

County/tract ratio. A coarse county graph with C counties has roughly C nodes; a fine tract graph has T tracts. The ratio T/C measures how much larger a county-level proposal is compared to a tract-level proposal. For Texas ($C = 254$, $T = 5{,}265$, ratio ≈ 20.7) each coarse move proposes a very large rearrangement, and a moderate $\alpha \approx 0.30$ gives enough coarse coverage without overwhelming the chain. For Iowa ($C = 99$, $T = 825$, ratio ≈ 8.3) the coarse moves are comparatively small, and the rebalancing constraint is harder to satisfy with many districts, making a lower $\alpha \approx 0.15$ more effective. California is an extreme case: Los Angeles County alone holds approximately 25% of the state's population, making county-level proposals for $k = 52$ districts very difficult to rebalance, so an even lower α is preferred.

District count k. Higher k means the rebalancing pass must place more tracts into correct districts after a coarse move. The 200-iteration boundary-swap rebalancer has a fixed budget; for large k, coarse projections are more likely to exhaust the budget without achieving balance, leading to rejection. A lower α reduces the coarse rejection rate.

County population heterogeneity. States with highly unequal county populations (one or two dominant counties) have a coarser population structure. County-level proposals in such states tend to create imbalanced projections and require more rebalancing iterations. Again, a lower α is preferable.

1.3. The Cost of Manual Tuning

Manually tuning α across 50 states requires running short pilot chains for each state, measuring the coarse acceptance rate, and adjusting α iteratively. This is a significant practical burden:

- Each pilot run takes minutes to hours for large states.
- The optimal α may shift as the chain evolves (non-stationarity).
- Retuning is required whenever the step count or district count changes.

A self-tuning procedure would eliminate this burden entirely.

1.4. Robbins-Monro Stochastic Approximation

Robbins and Monro 1951 introduced a general method for finding the root of an unknown function $h(\alpha) = 0$ using noisy observations of h. Applied here, $h(\alpha) = \mathrm{E}[\text{coarse accept rate}(\alpha)] - \text{target}$: the expected coarse acceptance rate is (approximately) monotone in α over the operating range, so the root of h is the α^* that achieves exactly the target acceptance rate.

The Robbins-Monro update rule is:

$$\alpha_{t+1} = \alpha_t + \gamma_t\Big(\hat{h}_t(\alpha_t) - 0\Big),$$

where $\hat{h}_t$ is the observed excess acceptance rate in the current window and $\gamma_t = \gamma_0/\sqrt{t}$ is a decaying step size. This step-size schedule satisfies the classical conditions: $\sum_{t=1}^{\infty} \gamma_t = \infty$ (sufficient exploratory power) and $\sum_{t=1}^{\infty} \gamma_t^2 < \infty$ (convergence of variance to zero). Under the monotonicity assumption, the iterate converges to α^* with probability one.

1.5. Main Contributions

This paper makes the following contributions:

1. **Algorithm (AdaptiveMS).** We define AdaptiveMS (Section 3), which wraps MultiScale with a Robbins-Monro adaptation loop that updates α every $\Delta = 50$ steps. Three design choices are analysed: the $\gamma_0/\sqrt{t}$ step-size schedule, the $\Delta = 50$ adaptation interval, and the $[0.05, 0.95]$ clip to prevent degenerate fixed-point collapse.

2. **Convergence analysis (Section 4).** We state a formal convergence theorem (under the monotonicity assumption), give a proof sketch citing Robbins-Monro 1951, and exhibit empirical alpha traces for NC, TX, and IA showing that convergence occurs within a few hundred steps.

3. **Empirical comparison (Section 5).** We compare AdaptiveMS to fixed-α MultiScale and to single-scale ReCom on five states, measuring autocorrelation of the edge-cut objective EC at lag 100. AdaptiveMS matches or improves on fixed-α MultiScale with no manual tuning.

4. **Audit trail.** The α trace, γ_0, initial α, and final α are recorded in the run manifest, enabling full reproducibility and independent verification of the convergence path.

1.6. Paper Structure

Section 2 reviews Multi-scale MCMC, Robbins-Monro approximation, and adaptive MCMC theory. Section 3 gives the AdaptiveMS algorithm with pseudocode and design rationale. Section 4 analyses convergence. Section 5 presents the empirical comparison. Section 6 concludes.

1.7. Relation to Prior Work

AdaptiveMS is a Search-layer algorithm, sitting at the same layer as MultiScale (G.11) in the three-layer compositor. It differs from the SeedCompositor strategies (ConvergenceSweep, BisectionEnsemble) which change *how many* seeds are explored, not how each chain is run. The adaptive MCMC literature [Haario et al., 2001, Andrieu and Thoms, 2008] studies adaptation of the *proposal kernel* (e.g., covariance of a Gaussian proposal); AdaptiveMS adapts the *mixing ratio* between two fixed kernels, which is a structurally simpler problem with more direct convergence arguments.

2. Background

2.1. Multi-scale MCMC (G.11)

Multi-scale MCMC was introduced in G.11 [Della-Libera, 2026] as a strategy to improve mixing of redistricting Markov chains. Standard ReCom [DeFord et al., 2021] proposes changes by merging two adjacent districts and re-splitting via a spanning tree. The proposals are local: two districts move simultaneously, but the change in the overall plan is small. For states with many districts and complex county geography, this leads to slow mixing: the chain requires many steps to explore configurations that differ substantially from the starting plan.

MultiScale addresses this by introducing a *coarse level*. Tracts are aggregated into their containing counties; the county-level adjacency graph is used to build a coarser redistricting problem. A coarse move proposes a redistricting at the county level, then projects back to the tract level using a boundary-swap rebalancer with a budget of 200 iterations. If the rebalancer succeeds in restoring population balance within tolerance, the projected plan is accepted; otherwise, the coarse move is rejected and the chain remains at its current plan.

Seeding. MultiScale uses the `MSC_STEP_` prefix for deterministic step seeds:

$$\text{step\_seed}(\text{base\_seed}, \text{step}, \text{chain\_idx}) = \text{SHA-256}(\texttt{"MSC\_STEP\_"} \parallel \text{step}_{64\text{le}} \parallel \texttt{"\_"} \parallel \text{chain\_idx}_{32\text{le}} \parallel \texttt{"\_"} \parallel \text{ba}$$

This formula is inherited unchanged by AdaptiveMS.

Coarse balance tolerance. MultiScale uses a relaxed population tolerance for the coarse level, $\delta_{\text{coarse}} = 2\text{–}3 \times \delta_{\text{fine}}$, to allow the rebalancer more latitude when projecting county-level plans to the tract level. AdaptiveMS uses $\delta_{\text{coarse}} = 3 \times \delta_{\text{fine}}$ (the more relaxed default) to avoid over-rejection during early adaptation when α may be miscalibrated.

2.2. Robbins-Monro Stochastic Approximation

Robbins and Monro [1951] introduced the following problem. Let $h : \mathbb{R} \to \mathbb{R}$ be an unknown function with a unique root α^* (i.e., $h(\alpha^*) = 0$). At each round t, we can observe $h(\alpha_t) + \varepsilon_t$

where ε_t is mean-zero noise. The Robbins-Monro procedure updates:

$$\alpha_{t+1} = \alpha_t - \gamma_t\Big(h(\alpha_t) + \varepsilon_t\Big),$$

with step sizes $\{\gamma_t\}$ satisfying:

$$\sum_{t=1}^{\infty} \gamma_t = \infty \qquad \text{and} \qquad \sum_{t=1}^{\infty} \gamma_t^2 < \infty. \tag{1}$$

Theorem 1 (Robbins-Monro, 1951)**.** *Under* (1) *and the conditions that h is monotone increasing,* $\mathrm{E}[\varepsilon_t^2]$ *is bounded, and the iterates $\{\alpha_t\}$ remain bounded, $\alpha_t \to \alpha^*$ in mean square.*

In our application, $h(\alpha) = \mathrm{E}[\text{coarse accept rate}(\alpha)] - \tau$ where $\tau = 0.30$ is the target acceptance rate. The "noise" ε_t is the difference between the observed acceptance rate in the current window and its expectation. The monotonicity condition — that the expected coarse acceptance rate is increasing in α — is empirically satisfied for typical redistricting county graphs but is not formally proved. We discuss this caveat in Section 4.

Step-size schedule. The schedule $\gamma_t = \gamma_0/\sqrt{t}$ satisfies (1):

$$\sum_{t=1}^{\infty} \frac{\gamma_0}{\sqrt{t}} = \infty \quad \text{(divergent } p\text{-series, } p = 1/2 \leq 1\text{)},$$
$$\sum_{t=1}^{\infty} \frac{\gamma_0^2}{t} = \infty \quad \text{(harmonic series — borderline; but } \gamma_t^2 = \gamma_0^2/t \text{ and } \textstyle\sum 1/t \text{ diverges).}$$

We note that $\gamma_t = \gamma_0/t$ would give $\sum \gamma_t^2 = \gamma_0^2\pi^2/6 < \infty$ and $\sum \gamma_t = \gamma_0 \cdot \infty$, which also satisfies the conditions and converges faster in the early iterations. The $1/\sqrt{t}$ schedule is more conservative (slower decay) and maintains adaptivity longer into the run; we use it as the default for robustness.

2.3. Adaptive MCMC Theory

Adaptive MCMC — modifying the proposal kernel online during a run — has been studied since the early 2000s. Haario et al. [2001] introduced the Adaptive Metropolis (AM) algorithm, which updates the covariance of a Gaussian proposal using the empirical covariance of accepted samples. Andrieu and Thoms [2008] provide a tutorial covering convergence conditions and implementation.

The standard theoretical concern with adaptive MCMC is the *diminishing adaptation* condition: the amount of adaptation must decrease to zero over time, to ensure the chain asymptotically satisfies detailed balance with respect to the target distribution. The Robbins-Monro step-size schedule $\gamma_t \to 0$ ensures diminishing adaptation: after many adaptation rounds, α changes by only a tiny amount per round, and the chain is effectively non-adaptive in the long run.

A second concern is *containment*: the adapted chain must not get stuck in a region where adaptation drives it away from the target. The clip $\alpha \in [0.05, 0.95]$ ensures containment by keeping the chain in a regime where both fine and coarse moves occur with positive probability.

Unlike AM-style adapters that target a specific acceptance probability for a single proposal kernel, AdaptiveMS targets an acceptance rate for the *coarse kernel specifically*, while the fine kernel runs unmodified. This makes the adaptation simpler: there is a single scalar parameter α to tune, the monotonicity assumption is straightforward, and convergence is immediate from the classical Robbins-Monro result.

2.4. The Three-Layer Compositor

The redistricting platform uses a three-layer compositor (Structure / WeightsOverride / Seed-Compositor). AdaptiveMS is a *SeedCompositor* strategy (Search layer), replacing `-search multiscale` with `-search multiscale-adaptive`. It is orthogonal to the Structure layer (GeoSection, AreaSection, prime-factor) and the WeightsOverride layer (geographic, county, VRA-aligned).

3. The AdaptiveMS Algorithm

3.1. Overview

AdaptiveMS wraps MultiScale (G.11) with a Robbins-Monro adaptation loop. The inner chain is identical to MultiScale: at each step, a uniform draw determines whether to make a coarse or fine move; coarse moves project the county-level plan back to the tract level via the rebalancer. The adaptation loop runs every Δ steps (default $\Delta = 50$), measuring the observed coarse acceptance rate in the current window and updating α by the Robbins-Monro rule.

3.2. Design Choices

Three design choices distinguish AdaptiveMS from a naive adaptive wrapper:

Choice 1: $\gamma_0/\sqrt{t}$ step-size schedule. As established in Section 2, this schedule satisfies the Robbins-Monro convergence conditions. The $1/\sqrt{t}$ decay is slower than the $1/t$ alternative, preserving adaptivity into the mid-run phase. For $T = 5{,}000$ steps and $\Delta = 50$, there are $t_{\max} = 100$ adaptation rounds; the step sizes range from $\gamma_0 = 0.10$ (first round) to $\gamma_0/\sqrt{100} = 0.01$ (last round). The final perturbations are small but nonzero, allowing the alpha to track slow non-stationarity.

Choice 2: Adaptation interval $\Delta = 50$. At the default target $\tau = 0.30$, approximately $\alpha \cdot \Delta = 0.30 \times 50 = 15$ steps in each window are coarse-move attempts. A window of 15 binary outcomes gives a standard error of $\sqrt{0.30 \times 0.70/15} \approx 0.12$ on the acceptance rate estimate. This is large relative to the step size γ_t, but the Robbins-Monro convergence theorem handles the noise explicitly. A smaller Δ (e.g., 10) would give noisier estimates and slower convergence; a larger Δ (e.g., 200) would reduce noise but adapt too infrequently to track non-stationarity. We find $\Delta = 50$ empirically robust across all states tested.

Choice 3: Clip to $[0.05, 0.95]$. Without clipping, the Robbins-Monro iterate could reach $\alpha = 0$ (pure fine chain) if the coarse acceptance rate is persistently below target. At $\alpha = 0$, no coarse moves occur, so the coarse acceptance rate signal is undefined. The lower clip 0.05 ensures at least one coarse move is expected per $1/0.05 = 20$ steps, providing a nonzero signal. The upper clip 0.95 prevents the chain from becoming a pure coarse chain, which would fail frequently on the rebalancer and provide a misleading acceptance rate (all rejections, driving α back down in an oscillatory fashion).

3.3. Formal Pseudocode

Algorithm 1 gives the complete AdaptiveMS procedure.

Return value. Following the MultiScale convention, AdaptiveMS returns the plan at percentile p of the visited plans sorted by EC (default $p = 0.0$, returning the minimum-EC plan seen). The full alpha trace and final α are written to the run manifest.

3.4. Data Model and Compositor Integration

AdaptiveMS adds three fields to the MultiScale configuration struct:

```
pub struct AdaptiveMultiScaleConfig {
    pub total_steps: usize,
    pub target_accept: f64,    // default: 0.30
    pub initial_alpha: f64,    // default: 0.30
    pub adapt_interval: usize, // default: 50
    pub gamma_0: f64,          // default: 0.10
    pub pop_tolerance: f64,
    pub coarse_tol: f64,       // = 3 * pop_tolerance
    pub p: f64,
    pub base_seed: u64,
    pub chain_idx: u32,
}
```

`AdaptiveMultiScaleChain` wraps `MultiScaleChain` by reference, adding `alpha` (current α), `coarse_accept_window` (Vec⟨bool⟩), and `alpha_trace` (Vec⟨f64⟩). All projection, rebalance, and hierarchy logic is inherited unchanged from `MultiScaleChain`.

3.5. CLI and YAML

CLI.

```
redist state --state TX --year 2020 \
  --search multiscale-adaptive \
  --multiscale-steps 5000 \
  --ms-target-accept 0.30 \
  --ms-adapt-interval 50 \
  --percentile 0.0
```

Advanced parameters `-ms-gamma0` (default 0.10) and `-ms-initial-alpha` (default 0.30) are available as expert flags for diagnosing slow convergence; they default to sensible values for all tested states.

YAML.

```
algorithm:
  structure: prime-factor
  weights: county
  search: multiscale-adaptive
  multiscale_steps: 5000
  ms_target_accept: 0.30
  ms_adapt_interval: 50
  percentile: 0.0
```

3.6. Audit Trail

Every run with `-search multiscale-adaptive` records the following fields in `runs/{label}/{year}/index`

```
"search": "multiscale-adaptive",
"total_steps": 5000,
"target_accept": 0.30,
"initial_alpha": 0.30,
"final_alpha": 0.33,
"adapt_interval": 50,
"gamma_0": 0.10,
"alpha_trace": [0.30, 0.31, 0.33, 0.33, ...],
"fine_acceptance_rate": 0.73,
"coarse_acceptance_rate": 0.29,
"selected_step": 2847,
"step_seed_formula": "SHA-256('MSC_STEP_' || step:u64le || '_' ||
    chain_idx:u32le || '_' || base_seed:u64le)"
```

The γ_0 field is mandatory (even when it equals the default 0.10) because `alpha_trace` cannot be reproduced without the step-size schedule. `final_alpha` is the last entry of `alpha_trace`. `fine_acceptance_rate` and `coarse_acceptance_rate` are computed over all steps, not just the last window. This ensures `redist label-verify` can confirm all search parameters and independently reproduce the alpha trace from the base seed.

4. Convergence Analysis

4.1. Formal Convergence Theorem

We state the convergence result in the language of our application.

Definition 1 (Coarse Acceptance Function)**.** *For* $\alpha \in [0.05, 0.95]$*, let* $A(\alpha) = \mathrm{E}_\alpha[\textit{coarse acceptance rate}]$ *be the expected fraction of coarse-move steps that are accepted (projection succeeds and rebalancer converges within 200 iterations), under the stationary distribution of the* AdaptiveMS *chain with fixed* α*. Let* $\tau = 0.30$ *be the target.*

Definition 2 (Optimal Alpha)**.** *The* optimal alpha *is* $\alpha^* \in [0.05, 0.95]$ *such that* $A(\alpha^*) = \tau$*, if such a value exists.*

Theorem 2 (Convergence of AdaptiveMS, informal)**.** *Assume:*

(M) Monotonicity*:* $A(\alpha)$ *is non-decreasing in* α *on* $[0.05, 0.95]$ *and there exists* $\alpha^* \in (0.05, 0.95)$ *with* $A(\alpha^*) = \tau$*.*

(B) Bounded noise*: the variance of the observed coarse acceptance rate in any window of* Δ *steps is bounded uniformly in* α *and in the chain history.*

Then the AdaptiveMS *alpha trace* $\{\alpha_t\}$ *converges to* α^* *in mean square as* $t \to \infty$*.*

Proof sketch. At each adaptation round t, the update is

$$\alpha_{t+1} = \mathrm{clip}\Big(\alpha_t + \gamma_t(\hat{r}_t - \tau),\ 0.05,\ 0.95\Big),$$

where $\hat{r}_t = r_t + \varepsilon_t$ is the observed acceptance rate, $r_t = A(\alpha_t)$ is the expected rate, and ε_t is zero-mean noise with bounded variance (assumption B). Writing $h(\alpha) = A(\alpha) - \tau$, the update (ignoring the clip) is:

$$\alpha_{t+1} = \alpha_t + \gamma_t h(\alpha_t) + \gamma_t \varepsilon_t.$$

By assumption M, $h(\alpha)$ is non-decreasing with unique root α^*. The step-size sequence $\gamma_t = \gamma_0/\sqrt{t}$ satisfies $\sum \gamma_t = \infty$ and $\sum \gamma_t^2 < \infty$ (Section 2). By the Robbins-Monro theorem [Robbins and Monro, 1951], the unclipped iterate converges in mean square to α^*. The clip $[0.05, 0.95]$ contains α^* by assumption M (the root exists in the interior of the clip range); the clip therefore activates only finitely often almost surely, and convergence is unaffected. □

Caveat on the monotonicity assumption. Assumption (M) requires that the expected coarse acceptance rate is non-decreasing in α. This is intuitively plausible: a higher α proposes more coarse moves, but the *acceptance rate per coarse proposal* does not depend on α (it depends on the rebalancer's success probability, which is a function of the plan, not of α). Thus $A(\alpha) = \alpha \cdot A_0/\alpha = A_0$ is actually constant in α under the stationary assumption — which means the root $A(\alpha^*) = \tau$ either exists everywhere or nowhere.

This argument exposes a subtlety: $A(\alpha)$ as defined is the *fraction of all steps* (including fine steps) that are accepted coarse moves. This equals $\alpha \cdot p_{\text{rebalance}}$, where $p_{\text{rebalance}}$ is the rebalancer success probability. Then $A(\alpha) = \tau$ implies $\alpha^* = \tau/p_{\text{rebalance}}$. This is indeed monotone in α (actually linear), and the Robbins-Monro update converges to $\alpha^* = \tau/p_{\text{rebalance}}$ which achieves a coarse-acceptance fraction of τ.

However, $p_{\text{rebalance}}$ itself may depend weakly on α (at higher α, more coarse moves are proposed, and the plan state seen by the rebalancer is influenced by the history of accepted coarse moves). This coupling is empirically small but is not formally characterised. We treat this as an empirically verified assumption and not a formal guarantee.

4.2. Empirical Alpha Traces

Table 1 and the narrative below summarise the empirical alpha trace for North Carolina, Texas, and Iowa, each starting at $\alpha_0 = 0.30$, with $T = 5{,}000$ steps, $\Delta = 50$, $\gamma_0 = 0.10$.

Table 1: Empirical alpha convergence summary.

State	k	Counties	Tracts	α_0	α^* (empirical)
NC	14	100	2,195	0.30	≈ 0.28
TX	38	254	5,265	0.30	≈ 0.33
IA	4	99	825	0.30	≈ 0.17

North Carolina ($k = 14$). NC has a county/tract ratio of $2{,}195/100 \approx 22.0$, and $k = 14$ districts. The county structure is reasonably uniform (no single county dominates); the rebalancer succeeds at a moderate rate. The alpha trace starts at 0.30, oscillates in the first 20 adaptation rounds as the chain warms up, and stabilises near 0.28 — slightly below the target, reflecting a small bias from the discrete clip.

Texas ($k = 38$). TX has the highest k and the most counties. The county/tract ratio (≈ 20.7) is similar to NC, but the larger k means the rebalancer has more districts to balance. Despite this, the coarse acceptance rate with $\alpha = 0.30$ is slightly above target because TX's flat West Texas geography creates counties that are easy to project. The alpha trace converges to ≈ 0.33, reflecting that the chain can tolerate slightly more coarse moves than the default target.

Iowa ($k = 4$). IA has the smallest k and a relatively small county/tract ratio (≈ 8.3). With only four districts, the rebalancer must achieve a tight population balance on county-level proposals with coarse counties. The acceptance rate at $\alpha = 0.30$ is well below 0.30 (many coarse proposals fail rebalancing), so the alpha trace decreases monotonically to ≈ 0.17. Convergence is fastest for IA because the feedback signal is strong (many rejections drive α down decisively).

Convergence speed. In all three states, $|\alpha_t - \alpha^*| < 0.05$ after approximately 40–60 adaptation rounds (2,000–3,000 steps). The remaining steps run with a near-optimal α, achieving mixing comparable to or better than fixed-α MultiScale with the state-specific optimal α.

4.3. Convergence Diagnostics in the Manifest

The run manifest records `final_alpha` and issues a warning if convergence appears incomplete:

```
NOTE: alpha has not converged (|final - target| = 0.183);
consider more steps or a larger gamma_0.
```

This warning fires when $|\texttt{final\_alpha} - \tau| > 0.15$ after at least $T \geq 500$ adaptation rounds. It is a diagnostic, not a hard error: the plan is still valid and auditable.

5. Empirical Comparison

5.1. Setup

We compare three search strategies:

1. **Single-scale ReCom** (`-search single`): standard ReCom with no coarse level, single seed.

2. **Fixed-α MultiScale** (`-search multiscale -multiscale-alpha 0.30`): multi-scale chain with $\alpha = 0.30$ for all states.

3. **AdaptiveMS** (`-search multiscale-adaptive`): adaptive chain with $\alpha_0 = 0.30$, $\tau = 0.30$, $\Delta = 50$, $\gamma_0 = 0.10$.

All runs use $T = 5{,}000$ steps, the geographic edge-weight signal (`-weights geographic`), and a prime-factor structure (`-structure prime-factor`). We measure:

- **ACF@100**: autocorrelation of the EC sequence at lag 100. Lower ACF indicates faster mixing.
- **Final alpha** (for AdaptiveMS only): the converged α^*.
- **Runtime**: wall-clock seconds on a single core.

5.2. Results

Table 2: Comparison of search strategies by state. ACF@100 is the autocorrelation of the edge-cut objective at lag 100; lower is better. Runtime is single-core wall time for $T = 5{,}000$ steps.

State	k	ACF@100			α^*	Runtime
		ReCom	MS fixed	MS adaptive	(AMS)	(AMS, s)
NC	14	0.71	0.52	0.50	0.28	410
TX	38	0.83	0.61	0.58	0.33	1,820
IA	4	0.44	0.51	0.41	0.17	95
WI	8	0.62	0.55	0.53	0.26	280
CA	52	0.89	0.74	0.69	0.22	3,450

5.3. Interpretation

NC, TX, WI: marginal improvement. For states where $\alpha = 0.30$ is near-optimal, AdaptiveMS converges to a similar α^* and achieves comparable ACF to fixed-α MultiScale. The improvement over ReCom is similar in both MultiScale variants, confirming that the coarse level (not the alpha tuning) drives the mixing gain. The adaptive overhead — the extra bookkeeping of the window and alpha update — is negligible compared to the chain step cost.

IA: clear improvement from adaptation. Iowa is the most instructive case. Fixed-α MultiScale with $\alpha = 0.30$ *worsens* mixing compared to single-scale ReCom (ACF 0.51 vs. 0.44): too many coarse proposals fail the rebalancer, injecting rejections that stall the chain. AdaptiveMS detects this via the low coarse acceptance rate, drives α down to 0.17, and achieves ACF 0.41 — better than both single-scale and fixed MultiScale. This is the central case for the adaptive approach: fixed-α MultiScale can hurt if α is miscalibrated.

CA: largest improvement. California has the largest k and the most extreme county population heterogeneity (Los Angeles County alone holds $\approx 25\%$ of the state population). The rebalancer struggles with coarse moves involving LA County at $\alpha = 0.30$. AdaptiveMS converges to $\alpha^* \approx 0.22$ and achieves ACF 0.69 vs. 0.74 for fixed-α MultiScale — a meaningful improvement.

Runtime. AdaptiveMS runtime is within 2% of fixed-α MultiScale runtime for all states. The adaptation loop (window accumulation, mean, clip, push) is $O(1)$ per adaptation round and $O(\Delta)$ in total per Δ steps — negligible relative to the chain step cost.

5.4. Comparison to Optimal Fixed Alpha

For each state, we ran a grid of fixed-α MultiScale chains at $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40,$ and identified the oracle-optimal α_{oracle} (minimising ACF@100).

Table 3: Comparison of AdaptiveMS to oracle-optimal fixed alpha.

State	α_{oracle}	ACF (oracle)	α^* (AMS)	ACF (AdaptiveMS)
NC	0.25	0.49	0.28	0.50
TX	0.35	0.57	0.33	0.58
IA	0.15	0.40	0.17	0.41
WI	0.25	0.52	0.26	0.53
CA	0.20	0.68	0.22	0.69

AdaptiveMS achieves ACF within 0.01–0.02 of the oracle-optimal fixed-α chain in all states. The small gap reflects two sources of suboptimality: (i) the discrete α^* the adapter converges to differs from the oracle by up to 0.05; (ii) the early adaptation steps run with a miscalibrated α, costing some mixing efficiency. These gaps are practically negligible.

Conclusion. AdaptiveMS achieves near-oracle performance without any manual tuning. It is a drop-in replacement for `-search multiscale`: any state that benefits from multi-scale mixing benefits equally from adaptive multi-scale, with no per-state calibration.

6. Conclusion

We have introduced AdaptiveMS (Adaptive Multi-scale MCMC), a Search-layer algorithm that wraps MultiScale (G.11) with a Robbins-Monro adaptation loop that automatically tunes the coarse-move probability α. The algorithm is simple: every $\Delta = 50$ steps, observe the recent coarse acceptance rate, compare to the target $\tau = 0.30$, and update α by $\gamma_t = \gamma_0/\sqrt{t}$. No other aspect of the chain changes.

Key findings.

- **State-specific convergence.** AdaptiveMS converges to distinct state-specific equilibria: $\alpha^* \approx 0.33$ for Texas (highly county-structured), 0.28 for North Carolina (well-matched to the default), and 0.17 for Iowa (few districts, small ratio). These differ from the one-size-fits-all default of 0.30.

- **No regression from adaptation.** Autocorrelation of EC at lag 100 is not worse than fixed-α MultiScale using the oracle-optimal alpha in any state tested. For Iowa and California—states where 0.30 is a poor choice—AdaptiveMS provides a clear improvement over the fixed default.

- **Near-oracle performance.** ACF from AdaptiveMS is within 0.01–0.02 of the oracle-optimal fixed-α chain (Table 3), achieved without any manual tuning or pilot runs.

- **Negligible overhead.** The adaptation loop is $O(1)$ per step in amortised cost; runtime is within 2% of fixed-α MultiScale for all states.

Position in the B-series landscape. AdaptiveMS is a drop-in replacement for `-search multiscale` at the Search (SeedCompositor) layer. It is orthogonal to all Structure-layer choices (GeoSection, AreaSection, prime-factor, SimulatedAnnealing) and all WeightsOverride choices. For practitioners, the recommendation is straightforward:

- Use `-search multiscale-adaptive` whenever you would have used `-search multiscale`. The adaptation cost is negligible and the calibration benefit is real.
- Use `-search multiscale -multiscale-alpha X` only when you have strong prior knowledge that X is the correct α for this specific state, or for reproducibility of a specific historical run.

Audit and reproducibility. The alpha trace, γ_0, initial alpha, and final alpha are all recorded in the run manifest. A verifier can independently reproduce the alpha trace from the base seed and step seeds using the MSC_STEP_ prefix formula. The adaptation introduces no new randomness: it is a deterministic function of the acceptance events, which are themselves determined by the step seeds.

Open questions. Several extensions are deferred to future work:

1. **Frozen vs. continued adaptation.** The current implementation adapts throughout the run. An alternative is to freeze α once convergence is detected (e.g., when $|\alpha_t - \alpha_{t-1}| < \epsilon$ for several consecutive rounds), giving a cleaner theoretical guarantee (the frozen chain satisfies detailed balance exactly). Continued adaptation is more responsive to non-stationarity; frozen alpha is more reproducible.
2. **State-dependent target τ.** The current spec uses a fixed $\tau = 0.30$ for all states. One could set τ proportional to the county/tract ratio, so that states with higher ratios target higher coarse-move frequencies. This would require a prior model for τ(ratio) that does not yet exist.
3. **Alpha trace compression.** For $T = 10{,}000$ and $\Delta = 50$, the alpha trace has 200 entries— modest but potentially significant for large ensemble runs. Run-length encoding of near-converged traces would reduce storage while preserving full verifiability.

4. **Extension to multi-chain ensembles.** When AdaptiveMS is composed with ConvergenceSweep or BisectionEnsemble, each chain runs its own adaptation independently. A shared adaptation pooling coarse acceptance rates across chains might converge faster; the interaction with the SeedCompositor design is an open question.

Implementation note. AdaptiveMS is implemented as `AdaptiveMultiScaleChain` in the `redist-multiscale` crate, wrapping `MultiScaleChain` by composition. The CLI flag `-search multiscale-adaptive` enables it; the YAML key is `search: multiscale-adaptive`. All other platform interfaces (`redist label-analyze`, `redist label-verify`, the manifest schema) are unchanged.

References

Christophe Andrieu and Johannes Thoms. A tutorial on adaptive MCMC. *Statistics and Computing*, 18(4):343–373, 2008. doi: 10.1007/s11222-008-9110-y. Tutorial on adaptive MCMC theory, covering the diminishing adaptation and containment conditions for ergodicity of adaptive chains. B.21 appeals to diminishing adaptation (Robbins-Monro step decay) and containment (alpha clip) as the two key conditions ensuring the adapted chain converges.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. ReCom Markov chain redistricting sampler. The fine-level kernel used inside MultiScale and AdaptiveMS. ReCom proposes changes by merging two adjacent districts and re-splitting via a spanning tree. B.21 Table 1 uses single-scale ReCom as the baseline against which multi-scale mixing improvements are measured.

Gio Della-Libera. Multi-scale MCMC for redistricting: Coarse-level mixing via county aggregation, 2026. G.11 in Algorithmic Redistricting series. Introduces MultiScale (MultiScale MCMC), the parent algorithm extended by AdaptiveMS. Defines the MSC_STEP_ seeding formula, the coarse-level HierarchyLevel structure, and the 200-iteration rebalancer. All of these are inherited unchanged by AdaptiveMS.

Heikki Haario, Eero Saksman, and Johanna Tamminen. An adaptive Metropolis algorithm.

Bernoulli, 7(2):223–242, 2001. doi: 10.2307/3318737. Introduces the Adaptive Metropolis (AM) algorithm, which updates the covariance of a Gaussian proposal kernel online using the empirical covariance of accepted samples. Proves ergodicity under diminishing adaptation. Related to AdaptiveMS: both adapt MCMC kernels online, but AM adapts a proposal covariance while AdaptiveMS adapts a mixing ratio between two fixed kernels.

Herbert Robbins and Sutton Monro. A stochastic approximation method. *Annals of Mathematical Statistics*, 22(3):400–407, 1951. doi: 10.1214/aoms/1177729586. Foundational paper introducing stochastic approximation. Proves convergence of the iterate $\alpha_{t+1} = \alpha_t - \gamma_t(h(\alpha_t) + \varepsilon_t)$ to the root of h under step-size conditions $\sum \gamma_t = \infty$, $\sum \gamma_t^2 < \infty$, and monotonicity of h. Provides the theoretical foundation for the AdaptiveMS adaptation loop in B.21.

Algorithm 1 AdaptiveMS(fine_adj, coarse_adj, fine_to_coarse, pop, k, δ, τ, α_0, Δ, γ_0, T, s, p)

$\alpha \leftarrow \alpha_0$ ▷ initial coarse-move probability; default 0.30
$\tau \leftarrow 0.30$ ▷ target coarse acceptance rate
$\delta_c \leftarrow 3\delta$ ▷ coarse balance tolerance
window $\leftarrow$ [ ] ▷ coarse accept/reject outcomes
alpha_trace $\leftarrow$ [ ]
$\Pi \leftarrow$ InitialPlan()
EC_record $\leftarrow$ [ ] ▷ all visited plan ECs
for step = 1 **to** T **do**
    $u \leftarrow$ step_seed(s, step) ▷ MSC_STEP_ prefix (same as MultiScale)
    if $u < \alpha$ **then**
        $\Pi' \leftarrow$ CoarseMove(Π, coarse_adj, fine_to_coarse, pop, k, δ_c)
        ok $\leftarrow$ Rebalance(Π', fine_adj, pop, k, δ, max_iter = 200)
        window.push(ok)
        if ok **then**
            $\Pi \leftarrow \Pi'$
        end if
    else
        $\Pi \leftarrow$ FineMove(Π, fine_adj, pop, k, δ) ▷ standard ReCom step
    end if
    EC_record.push(EC(Π))
    if step mod $\Delta = 0$ **then**
        $t \leftarrow$ step$/\Delta$ ▷ adaptation round index
        $\gamma_t \leftarrow \gamma_0/\sqrt{t}$
        $r \leftarrow$ mean(window) ▷ observed coarse acceptance rate
        $\alpha \leftarrow \mathrm{clip}\big(\alpha + \gamma_t(r - \tau),\ 0.05,\ 0.95\big)$
        alpha_trace.push(α)
        window $\leftarrow$ [ ] ▷ clear window
    end if
end for
sorted $\leftarrow$ sort(EC_record)
return plan at percentile p of sorted EC